

From: Pengwang Zhai pwzhai@umbc.edu
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG
Date: May 3, 2021 at 1:33 PM



To: Ibrahim, Amir (GSFC-6160) amir.ibrahim@nasa.gov, BAILEY, SEAN W. (GSFC-6160) sean.w.bailey@nasa.gov,
Franz, Bryan A. (GSFC-6160) bryan.a.franz@nasa.gov, Werdell, Jeremy (GSFC-6160) jeremy.werdell@nasa.gov,
Knobelspiesse, Kirk D. (GSFC-6160) kirk.d.knobelspiesse@nasa.gov,
Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] meng.gao@nasa.gov

Hello, All,

Here I deliver the second version of the radiative transfer package, which include the PACE Simulator, a new aerosol scattering matrix package, a new wrapper for calculating the aerosol reflectance table for atmospheric correction, and a number of scripts which help manage the workloads. The package can be downloaded via the following google drive link:

https://drive.google.com/drive/folders/1cUPr-yEAGdFNBE9gs_3Tw4X8QDREtvjC?usp=sharing

Please download and save all files to a location of your choice, let's call it RTSOS_PWZhai/.

Section 1: COMPILE

The steps to use it is: compile the package, configure the input files, and run the code with the executable file.

To compile, open a terminal and go to:

```
:$cd RTSOS_PWZhai/SOS_Callable/compile
```

then try to compile the monochromatic code:

```
:$make
```

If this is good, then edit the PACE Simulator make file to set the hdf5 library paths:

```
:$emacs makefile_PACE_Simulator_DoubleK
```

In the make file you need to specify the hdf5 library paths:

```

HDF5DIR = /usr/local/Cellar/hdf5@1.10/1.10.7
HDF5LIB=-I$(HDF5DIR)/include
H5FC=$(HDF5DIR)/bin/h5fc
LIBSHDF= $(HDF5LIB) -L$(HDF5DIR)/lib/ -lhdf5 -
lhdf5_fortran

```

If these are correctly set, you can compile the pace simulator:

```
:$make -f makefile_PACE_Simulator_DoubleK
```

This will compile the pace simulator.

To compile the aerosol LUT wrapper, do the following:

```
:$make -f makefile_GSFC_AC_LUT
```

Of course, you will need to specify the hdf5 library paths in the makefile first.

Section 2: RUN the MODELS

Section 2.1 PACE Simulator

Next we may try out the pace simulator:

```

:$cp rtsos_PACE_Simulator_DoubleK.exe
../test/PACE_Simulator/
:$cd ../test/PACE_Simulator/

```

Check out “notes_to_run_pace_simulator” for the use of pace simulator, which is located in RTSOS_PWZhai/SOS_Callable/test/PACE_Simulator/. In the same folder you may also find a folder “python_script” which contains three python scripts including “pace_simulator_inputfile_prepare.py”, which can be used to generate a batch of pace simulator input files.

Current version can conveniently generate synthetic datasets for OCT, HARP, and SDEV for flexible

datasets for UCI, NARR, and SPCAN for flexible atmospheric and ocean condition. The input parameters are specified in input files.

Right now the aerosol models are: IAEROSOL=1-20, where IAEROSOL=1-10 is the old Shettle&Fenn model; and IAEROSOL=11-20 is Zia's new aerosol model. IRH=1-8 where $RH(IRH) = (/0.30, 0.50, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95/)$.

In this version, we have also provided a single scattering matrix package, which can be used to calculate the single scattering matrix for a flexible composition of aerosols, including fine and coarse mode dusts, water soluble, biomass burning, etc. To use this feature, IAEROSOL=-99 need to be set, and a single scattering matrix file name need to be provided to the pace simulator via the input file. See the python script for details.

Section 2.2 Single Scattering Package

To use the single scattering matrix package, go to:

```
:$cd RTSOS_PWZhai/SOS_Callable/compile
:$make -f makefile_aerosol_phmx_cal
```

You will also need to specify the HDF5 library path in the make file. After compiling, cp the exe file to the test location:

```
:$cp Aerosol_Phmx_Cal.exe
../test/PACE_Simulator/Aerosol_PhaseMatrix_Cal/
```

The python script to prepare the input file for Aerosol_Phmx_Cal.exe is located in RTSOS_PWZhai/SOS_Callable/test/PACE_Simulator/python_script/Aerosol_PhaseMatrix_Cal_InputPre.py. To run it, go the folder:

```
:$cd ../test/PACE_Simulator/python_script
```

```
:$cp Aerosol_PhaseMatrix_Cal_InputPre.py
../Aerosol_PhaseMatrix_Cal/
:$python Aerosol_PhaseMatrix_Cal_InputPre.py
```

Here you can look for what input file has been generated. One you find it, say input_nmode3, do the following to generate the single scattering file:

```
:$cd ../Aerosol_PhaseMatrix_Cal
:$Aerosol_Phmx_Cal.exe input_nmode3
```

Section 2.3 AEROSOL LUT Wrapper

In the compilation section, we have compiled the wrapper. Now let's copy it to the test section to try it out:

```
:$cd RTSOS_PWZhai/SOS_Callable/compile/
:$cp rtsos_GSFC_AC_LUT.exe ../test/test_MODIS_LUT/
:$cd ../test/test_MODIS_LUT/python_scripts_inputfiles/
:$ls *txt
```

You will see

```
MODIS_MIE_DIR.txt
OCI_MIE_DIR.txt
SEAWIFS_MIE_DIR.txt
aerosol_LUT_datafield_explain.txt
```

Here you need to open all three txt files: MODIS_MIE_DIR.txt, OCI_MIE_DIR.txt, SEAWIFS_MIE_DIR.txt, where the wrapper will read in the directories for the Mie single scattering files. For example, you may open MODIS_MIE_DIR.txt, and change the line to be:

```
RTSOS_PWZhai/SOS_Callable/MODIS_Mie_Database
```

Moreover you need to specify the auxiliary folder:

```
:$emacs auxiliary_directory
```

and change the line to be:

```
RTSOS_PWZhai/SOS_Callable/Data/
```

Now run the python script to prepare the input files for the wrapper:

```
:$python rt_GSFC_LUT_pre.py
```

You will be prompted to enter the instrument label: instrument_label=?, enter 1 for OCI, 2 for MODIS and 3 for SeaWifs:

Just enter a number for which instrument you want to simulate.

After python scripts finished processing, you will see a total of 80 folders are created under the current directory. Each directory corresponds to one aerosol model. To run all the case, one way is to use the sh script: "caserun.sh" which can be found in test/test_MODIS_LUT/python_scripts_inputfiles/. Before you run, take a look at the script:

```
:$less caserun.sh
```

```
#!/bin/bash
execfile=../../rtsos_GSFC_AC_LUT.exe
for casepath in ./Inputfile_Dir_OCI*/ ;
do
    cd "$casepath"
    find . -name "input_*" | parallel -I% --max-args 1
    -j18 $execfile %
    cd ..
done
```

You need to modify the string "Inputfile_Dir_OCI*/" to be appropriate format. For OCI, no need to change; for MODIS, change OCI to MODIS; for SeaWifs, change OCI to

MODIS, change UCI to MODIS; for SeaWiFS, change UCI to SeaWiFS, etc.

The option `-j18` is used to specify the number of simultaneous parallel jobs run on the computer nodes. Change 18 to a number which matches the computer specification.

To generate the aerosol LUT, run:

```
:$caserun.sh > logfile &
```

This will take a while to finish. The current setting is not for the most efficient simulation. If time is an issue, let me know and I will explore a more efficient setting while keeping the accuracy.

If everything goes well, you should get all 80 aerosol model simulated. To post process the data, use “AC_GSFC_LUT_Data_Process_withNumba.py” which can be found in
test/test_MODIS_LUT/python_scripts_inputfiles/:

```
:$python AC_GSFC_LUT_Data_Process_withNumba.py
```

You will need to input the same instrument label you used to generate the radiative transfer dataset.

The final post-processed dataset will be found in each folder that contains the datasets. For example, you may check one folder:

```
:$cd Inputfile_Dir_SeaWifsAerosolModel11rh_0.300000/  
:$ls rt_sos_aerosol*
```

The file is:

```
rt_sos_aerosolSeaWifs_0.300000f0.000000.h5
```

I am trying to write what I can on how to use and simulate the aerosol LUT. However, I realize that

there must be tons of questions on the to deal with a number of exceptions. Please let me know and I will respond. It might be better for us to meet and I will give a demo on the usage of the model.

Cheers,

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 29, 2021, at 12:15 PM, Gao, Meng (GSFC-616.0)
[SCIENCE SYSTEMS AND APPLICATIONS INC]
<meng.gao@nasa.gov> wrote:

Hi Amir and Pengwang, nice to see all the good agreements!

Just FYI, there is already a repo in gitlab for Pengwang's PACE simulator code called Zhai_RT. Maybe Sean and Don can help to merge the new code together.

Best,

Meng

From: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Sent: Thursday, April 29, 2021 11:58 AM

To: Pengwang Zhai <pwzhai@umbc.edu>

Cc: Ahmad. Ziauddin (GSFC-616.0)[Science Application

International Corp.] <ziauddin.ahmad-1@nasa.gov>;
BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>;
Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>;
Werdell, Jeremy (GSFC-6160)
<jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D.
(GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao,
Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS
INC] <meng.gao@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPB

Hi Pengwang,

I agree the results looks good and it's more ready
for delivery. I have made a comparison of the F11 and
SSA for both aerosol models and they look in
agreement.

<Outlook-subblhd0.png>

The SSA for both models looks pretty much the same as
in the LUTs. I noticed that LUT/AFRT F11 for r30 f00
model is slightly higher close to 125 deg scattering
angle. Very minor differences and I think overall
things looks great!!

Regarding Version Control we typically use git. We
have Gitlab for our processing system. I think we can
create a repo there for the RT code and the wrapper
and provide you access to it so we can centralize our
code changes and track them for further development.
Sean and Don are the gate keepers to Git stuff. Sean
is on vacation so I will speak with Don to see what
the best way is to do this.

Regards,
Amir

From: Pengwang Zhai <pwzhai@umbc.edu>

Sent: Thursday, April 20, 2021 10:51 AM

SENT: Thursday, April 29, 2021 10:51 AM
To: Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>;
BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>;
Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>;
Werdell, Jeremy (GSFC-6160)
<jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D.
(GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao,
Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS
INC] <meng.gao@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Yes, Amir, that is exactly the reason. For atmosphere with aerosols, the diffuse light would be larger, but the direct glint would be attenuated more. This two factors contradict with each other in opposite direction. For some aerosol loading, the difference would be negative (glint appears darker), whereas in some other cases the difference would be larger (reflectance is larger with aerosol loading).

At this point, because you confirm that the AFRT results are only for flat surface, the Rayleigh comparison is no longer needed, as we model different things. However, it would be still good to check the scattering matrix and ssa, especially for the case with $rh=0.30$, which I newly implemented.

Things are looking good. I will think about how to deliver it. Version control is important, as you suggested. I would listen from you on how we build a mechanism so that we can share more easily and I can upgrade the code with you folks easier.

In my latest version, I made the simulation process the same for all three instruments, SEAWIFS, MODIS, and OCI. More to come after I have a better idea of how I deliver.

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 29, 2021, at 10:33 AM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Thanks Pengwang!

Thinking about it again, I realize that since our aerosol LUTs are calculated only for flat surface, we should expect a difference for $\sigma > 0$. Also, what I am trying to understand is this figure below where when you are close to the principal plane, the total (rayleigh+aerosol+glint) - (rayleigh+glint) shows negative reflectance. I think with higher aerosol optical depth, the glint is more attenuated, thus the glint is reduced at toa for the total signal. That brings up the question of how to separate the aerosol in that case then. If we will stick to the operational AC method, I think we are good as things match well, but I am just trying to understand this part.

<image.png>

I will also do the other comparisons you mentioned and will get back to you.

Cheers,
Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Thursday, April 29, 2021 9:10 AM
To: Ibrahim, Amir (GSFC-6160)
<amir.ibrahim@nasa.gov>
Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>;
BAILEY, SEAN W. (GSFC-6160)
<sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>;
Knobelspiesse, Kirk D. (GSFC-6160)
<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]
<meng.gao@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPB

Hello, Amir,

Thanks for the plots, which actually looks good to me, for the non-glint viewing angles.

To resolve the difference in glint angles, here is a few steps that we can do to narrow down the scope:

1. Would you compare the reflectance including the glint region for the pure Rayleigh case (aerosol optical depth =0) with a sigma value > 0? This is to confirm the BRDF model in both AFRT and my SOS code are consistent. Ideally, we would do two wavelengths, one short and one long.

2. Would you also compare my scattering matrix and aerosol single scattering albedo (SSA) to Zia's? This step we can eliminate the possibility of single scattering matrix differences.

3. If both #1 and #2 comparison is good, I can rerun one case using my Monte Carlo method, which is completely independent, and should provide a third option benchmark for "truing" the results in the

option benchmark for verifying the results in the glint.

4. Of course, it would be great if Zia can confirm the steps to reconstruct the total reflectance in the glint for AFRT.

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 28, 2021, at 4:51 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

I compared the second aerosol model you provided ($f=80, r=70$) and the agreement (qualitatively) looks great for most geometries. However, for some geometries close to the direct glint and when $\sigma > 0$, I see differences. The aerosol reflectance is smaller than AFRT. Again, I subtracted the total reflectance (i.e. aerosol+rayleigh+glint) from the rayleigh+glint case. I am not sure of the cause. Zia, would you please confirm that this is how we calculate the aerosol reflectance by doing this subtraction.

Also, please let me know if you wish not to receive emails from this discussion. I hope I am not

spamming anyone.

Cheers,

Amir

From: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Sent: Tuesday, April 27, 2021 4:35 PM

To: Pengwang Zhai <pwzhai@umbc.edu>

Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>;

BAILEY, SEAN W. (GSFC-6160)

<sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-

6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy

(GSFC-6160) <jeremy.werdell@nasa.gov>;

Knobelspiesse, Kirk D. (GSFC-6160)

<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-

616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]

<meng.gao@nasa.gov>

Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Thank you Pengwang. I will test that set and will
get back to you.

I appreciate all the efforts.

Regards,

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>

Sent: Tuesday, April 27, 2021 12:12 PM

To: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>;

BAILEY, SEAN W. (GSFC-6160)

<sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-

6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy

(GSFC-6160) <jeremy.werdell@nasa.gov>;

Knobelspiesse, Kirk D. (GSFC-6160)

<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-

616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]

<meng.gao@nasa.gov>

Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Hello, Amir,

I have updated the simulation with all the
identified changes implemented. The new dataset can
be downloaded through the following google drive
link:

https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fdrive.google.com%2Fdrive%2Ffolders%2F1pk331jfxmA47_051T0Trr799iRW4Irb%3Fusp%3Dsharing&data=04%7C01%7Camir.ibrahim%40nasa.gov%7C523375aaf52e465bd61508d90b1e5b03%7C7005d45845be48ae8140d43da96dd17b%7C0%7C0%7C637553047314064297%7CUnknown%7CTWFpbGZsb3d8eyJWIjoiMC4wLjAwMDAiLCJQIjoiV2luMzIiLCJBTiI6Iik1haWwiLCJXVCI6Mn0%3D%7C1000&sdata=nbyUaBVwzwrTLzMJ2114dAv18GgGgMrjyAMNZXbdbfs%3D&reserved=0

In addition to the previous case of $f=0$ and $rh=0.3$,
a new case is added: $f=0.8$ and $rh=0.7$. You may find
two h5 files in the link.

I have also reported surface glint contribution to
the TOA reflectance in the file.

The new database uses reference wavelength of 865 nm
for SeaWifs, as you have advised.

Whitecap has been turned off.

Two modes of surface roughness parameterizations are
implemented. One is Cox&Munk 1954, the other is
Gordon & Wang 1992. This current two dataset uses
GW1992.

Let me know if I have missed something.

Yours

TOURS,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 20, 2021, at 5:40 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

I converted the wavelengths, but the interpolation might not be accurate. The reference wavelengths vary depending on the instrument, so for SeaWiFS it's 865 nm. Also, I think another aerosol model might be useful such as Rh=70, volume fmf=80?

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Tuesday, April 20, 2021 5:02 PM
To: Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG

Hello, Amir,

Thanks for sharing the plot. I guess the conversion between 870 to 550 is something to be blamed for the differences. I will redo the simulation with reference wavelength at 870 nm and send you the new data for comparison.

By the way, are you using 870 nm as the reference wavelength for all instruments? Or you use slightly different reference wavelengths for different instruments?

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 20, 2021, at 4:33 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

Assuming I am reading the correct aerosol model and converting the aot reference from 870 to 550 nm correctly, I see some discrepancies between the AFTR and SOS. In the attachment, I am plotting the aerosol reflectance by subtracting the TOA reflectance for aot=0 from the total at aot=0.1 at 550 nm. I am not sure if this difference has

something to do with the fact that the model we are comparing is at the extreme end of the table (i.e. $R_h=30$, $f_{mf}=0$).

Cheers,

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>

Sent: Tuesday, April 20, 2021 11:28 AM

To: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Cc: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>; BAILEY, SEAN W. (GSFC-6160)

<sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>;

Knobelspiesse, Kirk D. (GSFC-6160)

<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]

<meng.gao@nasa.gov>

Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG

Thanks, Amir, for the info. I will revise the input file so that the white cap can be turn on and off. Also I will report the glint contribution separately.

Once you are comfortable the current simulation, I can finalize the simulation package and deliver it together with the next version of the PACE simulator. Since the difference you see mostly in the large sigma value cases, would you check the small sigma cases with aerosols so that we have some common ground?

Take care,

Yours,

Pengwang

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 20, 2021, at 10:47 AM, Ibrahim, Amir
(GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

I have clarified some detail with Zia and others regarding the whitecaps. We actually don't simulate whitecaps in the LUTs as it is treated separately in the atmospheric correction algorithm similar to the glint. However, the glint is simulated of course but subtracted in a post processing step. Thus, no need to run whitecaps. Our vision for the next step is to include both glint and whitecaps in the LUTs directly so having the ability to turn on or off the whitecaps in the RT would be useful.

Thanks,
Amir

From: Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>

Sent: Saturday, April 17, 2021 1:52 PM

To: Pengwang Zhai <pwzhai@umbc.edu>

Cc: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>; Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell,

Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>;
Knobelspiesse, Kirk D. (GSFC-6160)
<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-
616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]
<meng.gao@nasa.gov>; Ahmad, Ziauddin (GSFC-616.0)
[Science Application International Corp.]
<ziauddin.ahmad-1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Hi Pengwang,
In the current LUTs for Modis, SeaWiFS etc., the
glint correction includes attenuation of light
by Rayleigh and aerosol optical thickness of the
model atmosphere.
Zia

On 4/16/21, 4:46 PM, "Pengwang Zhai"
<pwzhai@umbc.edu> wrote:

Hello, Zia, and all,

After reviewing your code, I found that your
white cap fraction calculation is the same as
what I had before, i.e.,

```
      ! FWC=2.95E-6*WNDSPD**3.52D0      ! EQUATION (3),  
FRASER, JGR, 1997.  
      ! WCLBD0=0.22D0                      ! VALUE  
OBTAINED IN FRASER, JGR, 1997.
```

Though I assumed albedo=0.22 and you have a
spectral dependence.

Now I have revised my code according to Sean's
suggestion:

```
      IF(WNDSPD<6.33D0)THEN  
          FWC=0.0D0  
      ELSE IF (WNDSPD<12.0D0)THEN  
          FWC=8.75E-5*(WNDSPD-6.33D0)**3
```

```
ELSE IF(WNDSPD>12.0D0)THEN
      FWC=8.75E-5*(12.0D0-6.33D0)**3
ENDIF
WCLBDO=WHITECAP_REFLECTANCE(WAVELENGTH) !
WAVELENGTH in micron
```

I am fine with either options, but just need a definite instruction on which option to use in calculating the LUT for atmospheric correction.

The other question was on the glint separation. Amir asked me whether I can provide the glint separately. Before I do anything, I asked whether the glint used in the AC LUT is just the bidirectional reflectance of the surface of the sun light, or it has also been attenuated by Molecules and Aerosols. This is a simple thing to calculate. But I want to give the exact same thing which is needed in the processing algorithm.

Cheers,

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 16, 2021, at 2:54 PM, Ahmad, Ziauddin
(CSEC 616 0) [Science Application International]

(GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov> wrote:

Amir,

The attached file show how the foam calculations are handled RT2 module of AFRT. I do not understand your question re: the glint separation and white caps reflectance.

Zia

From: "Ibrahim, Amir (GSFC-6160)"

<amir.ibrahim@nasa.gov>

Date: Thursday, April 15, 2021 at 2:26 PM

To: Sean Bailey <sean.w.bailey@nasa.gov>,

Pengwang Zhai <pwzhai@umbc.edu>

Cc: Bryan Franz <bryan.a.franz@nasa.gov>,

"Werdell, Jeremy (GSFC-6160)"

<jeremy.werdell@nasa.gov>, "Knobelspiesse, Kirk

D. (GSFC-6160)" <kirk.d.knobelspiesse@nasa.gov>,

"Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND

APPLICATIONS INC]" <meng.gao@nasa.gov>, "Ahmad,

Ziauddin (GSFC-616.0)[Science Application

International Corp.]" <ziauddin.ahmad-

1@nasa.gov>

Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

I agree with Sean's suggestion, but I would like to get some feedback from Robert and Heidi on the topic and see if there is a possible updated model for OCI to be used. For now, I suggest sticking to what we do operationally and over the year we will revise every aspect of the model.

Zia, I am still waiting for your input on the glint separation part.

Thanks,

Amir

From: BAILEY, SEAN W. (GSFC-6160)

<sean.w.bailey@nasa.gov>
Sent: Thursday, April 15, 2021 2:02 PM
To: Pengwang Zhai <pwzhai@umbc.edu>
Cc: Ibrahim, Amir (GSFC-6160)
<amir.ibrahim@nasa.gov>; Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>; Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG

Good question __

One might say why do our tables go to 30m/s in the first place...but since they do (at least for now), I see your point.

Using Koepke outside the valid range for Stramska would introduce a discontinuity. If the goal is to be consistent with the retrieval processing, then I would say use zero below 6.33 m/s and fix the value at 12 m/s for windspeeds above 12 m/s.

...but that's kinda ugly...

It's not so awful, though. Whitecaps aren't really that important in our retrievals. I know others would argue differently, but for a fully developed sea at 12 m/s Stramska gives us a reflectance of about 0.003...

Don't let my word be the last on the topic...anyone else with a thought should chime in __

Sean

On 4/15/21, 12:31 PM, "Pengwang Zhai"
<pwzhai@umbc.edu> wrote:

Thanks, Sean, for clarifying this, which is helpful for me to understand the current implementation.

I will still need to put in something for wind speed larger than 12 m/s, as the LUT parameters include wind speed up to some 30 m/s. Should I use Koepke (1985) for wind speed smaller than 6.33 m/s and larger than 12 m/s?

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 15, 2021, at 11:56 AM, BAILEY, SEAN W.
(GSFC-6160) <sean.w.bailey@nasa.gov> wrote:

Pengwang,

In our processing we limit the windspeed for the whitecap correction to a max of 12m/s. We rarely see clear skies with wind speeds above 12m/s, so it's not really a problem. We did an analysis of this over a decade ago.

analysis of this over a decade ago:

<https://oceancolor.gsfc.nasa.gov/reprocessing/r2009/whitecap/>

Regards,
Sean

From: Pengwang Zhai <pwzhai@umbc.edu>
Date: Thursday, April 15, 2021 at 11:50 AM
To: "Ibrahim, Amir (GSFC-6160)" <amir.ibrahim@nasa.gov>
Cc: "Franz, Bryan A. (GSFC-6160)" <bryan.a.franz@nasa.gov>, "Werdell, Jeremy (GSFC-6160)" <jeremy.werdell@nasa.gov>, "Knobelspiesse, Kirk D. (GSFC-6160)" <kirk.d.knobelspiesse@nasa.gov>, "Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC]" <meng.gao@nasa.gov>, "BAILEY, SEAN W. (GSFC-6160)" <sean.w.bailey@nasa.gov>, "Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.]" <ziauddin.ahmad-1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG

Hello, Amir,

I checked the white cap fraction parameterization, see attached screenshot:

<image001.png>

Note that Stramska and Petelski (2003) version increases larger than 1 for wind speed approaches 30. I read the ocean optics info, which says the validity range of Stramska and Petelski (2003) is between 6.33 and 12 m/s.

What would you deal with wind speed larger than 12 m/s? Should be use Koepke (1985) for wind

speed smaller than 6.33 and larger than 12 m/s?
This would create a discontinuity on FWC
values.

Cheers,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 8, 2021, at 11:14 AM, Ibrahim, Amir
(GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

To clarify things, the diffuse transmittance
of the atmosphere to take the Lw to TOA is the
Rad_DiffT_Boa2Toa?

Regarding the whitecaps, we use Eq. (4) in the
link for undeveloped seas and $WS > 6.33$ m/s. So
that could also cause a difference.

I am not sure if we need the surface BRDF only
or not. I believe it's the surface only since
in the L2 processing we add the glint back by
using the glint reflected radiance attenuated
by the Rayleigh and aerosols to TOA. Zia can
you clarify to us this part.

Thanks.

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Tuesday, April 6, 2021 9:41 AM
To: Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>; Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer integration into OBPG

Hello, Amir,

I am glad that the table is understandable. My answers to your questions is below:

• What is Irrad_Transmittance_Toa2Boa and Rad_DiffT_Boa2Toa?

Irrad means irradiance; Toa means Top of atmosphere; Boa means Bottom of atmosphere; Rad means radiance; DiffT means diffuse transmittance. In summary:

Irrad_Transmittance_Toa2Boa: irradiance transmittance from TOA to BOA.

Rad_DiffT_Boa2Toa: radiance diffuse transmittance from BOA to TOA.

• The optical depth in AFRT simulations is typically done for the long MTP

is typically done for the long NIR wavelengths of ocean color sensors (i.e., ~870nm). Your tables are at 550 nm, which is not a problem, but I am letting you know.

If the optical depth values you provided are referred to the reference wavelength of 870 nm, I will need to revise the code to be consistent with. Currently I assumed 550 nm.

- Can you remind me what you assumed for whitecaps?

For Whitecaps, this is what I am using:

```
FWC=2.95E-6*WINDSPD**3.52D0    ! EQUATION (3),  
FRASER, JGR, 1997.  
WCLBD0=0.22D0                  ! VALUE OBTAINED  
IN FRASER, JGR, 1997.
```

where FWC is the fraction of white cap, and WCLBD0 is the white cap reflectance.

This could be a source of the differences that you observed for large sigma, as I am using a different wind speed grid to match your sigma values. However, you are using different parameterization of sigma with different wind speed, which may impact the white cap. If this is the source, I can certainly change this by parameterize FWC in terms of sigma instead of wind speed.

- We would need to separate out the surface contributions similar to the operational tables. Do you save the surface reflection matrix?

I currently do not save the direct surface reflection matrix. However, this can be done

by post processing. When you say surface reflection matrix, do you mean the rough surface reflection BRDF only? Or you also include the attenuation by the Rayleigh and aerosols? I just need to know what you need in order to provide the correct numbers.

If you share the difference figures that you have, I will see if I can guess the reasons of the differences.

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 5, 2021, at 6:23 PM, Ibrahim, Amir
(GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

Thanks again for providing this table. It was really useful and easy to understand. I did a quick benchmark comparison on the Rayleigh scattering and the agreement between our LUT and your outputs looks good. I noticed bigger differences, however, in the glint and when sigma is too high. I am not comparing AFRT output directly, however, I am reconstructing our LUTs to get back the TOA reflectance for

the Rayleigh case (i.e., adding back the direct glint and whitecaps).

The difference in the glint is probably because of the glint reflectance calculations (I will review that later). I also see consistent difference at sensor angle > 80. I am not looking at an exact match at this point and that's why I didn't do a percent difference figures.

Can you clarify few of points below?

- What is Irrad_Transmittance_Toa2Boa and Rad_DiffT_Boa2Toa?
- The optical depth in AFRT simulations is typically done for the long NIR wavelengths of ocean color sensors (i.e., ~870nm). Your tables are at 550 nm, which is not a problem, but I am letting you know.
- Can you remind me what you assumed for whitecaps?
- We would need to separate out the surface contributions similar to the operational tables. Do you save the surface reflection matrix?

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Thursday, April 1, 2021 3:54 PM
To: Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>; Ahmad, Ziauddin (GSFC-616.0)[Science Application

Ziauddin (GSFC-010.0) [Science Application
International Corp.] <ziauddin.ahmad-
1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Hi, Amir,

Let's try dropbox:

<https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.dropbox.com%2Fsh%2Fwp1z5a4iatbcpi4%2FAAB6Ut8TGDJTdwkHEkyx0s30a%3Fdl%3D0&data=04%7C01%7Camir.ibrahim%40nasa.gov%7C523375aaf52e465bd61508d90b1e5b03%7C7005d45845be48ae8140d43da96dd17b%7C0%7C0%7C637553047314064297%7CUnknown%7CTWFpbGZsb3d8eyJWIjoiMC4wLjAwMDAiLCJQIjoiV2luMzIiLCJBTiI6IkJ1aWwiLCJXVCI6Mn0%3D%7C1000&sdata=1DbCKUMvokgIlmxQmQglGyj9iB6ShuFFJg0d%2Br3gq4I%3D&reserved=0>

For the diffuse transmittance, I agree that we can just use BOA to TOA.

Let me know what you find.

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department, UMBC
pwzhai@umbc.edu

On Apr 1, 2021, at 3:32 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

Thanks for the files but I can't download them for some reason. Regarding the T00/TOA transmittance, T00 needs to be recorded at BOA (i.e. after applying the transmission/reflection matrix of the surface), then calculated the transmittance from that BOA signal. I really think that the BOA/TOA transmittance is fine since our Lambertian assumption is not accurate in any case.

Once I have access, I will compare these values to the standard tables.

Regards,
Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Thursday, April 1, 2021 2:52 PM
To: Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0)[SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>; Ahmad, Ziauddin (GSFC-616.0)[Science Application International Corp.] <ziauddin.ahmad-1@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer

integration into OBPG

Hello, Amir and Bryan, and all,

I have simulated a set of data for SeaWifs with one aerosol model (fine mode fraction: 0; relative humidity = 0.3). The files can be downloaded in the following google drive link:

https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fdrive.google.com%2Fdrive%2Ffolders%2F1pk331jffjxmA47_051T0Trr799iRW4Irb%3Fusp%3Dsharing&data=04%7C01%7Camir.ibr%7C523375aaf52e465bd61508d90b1e5b03%7C7005d45845be48ae8140d43da96dd17b%7C0%7C0%7C637553047314064297%7CUnknown%7CTWFpbGZsb3d8eyJWIjo%3D%7C1000&sdata=nbyUaBVwzwrTLzMJ2114dAv18GgGgMrjyAMNZXbdbfs%3D&reserved=0

In the file you may find three files:

rt_sos_aerosolSeaWifsr_0.300000f0.000000.h5
aerosol_LUT_datafield_explain.txt
MODIS_LUT_Data_Check.ipynb

The h5 file is the actual datafile, 'aerosol_LUT_datafield_explain.txt' is the data file structure with dimensions explained. The ipynb file reads in and plots some of the data.

Please check the file out and let me know how does this compare to the LUT data you are currently using. Also let me know anything that I am missing in the file.

Previously we discussed whether to put the

Previously we discussed whether to put the sensor under the water surface when we calculate the radiance diffuse transmittance. The email thread was not finished so far so I just calculated both: from top of the ocean (T00) to T0A and from bottom of the atmosphere (BOA) to T0A. From the plot, it is obvious that we need to use BOA to T0A, and the T00 to T0A values are merely around 0.5, which is primarily due to the n^2 laws. However, I leave that decision up to you.

I did this in a short amount of time, so I may have some mistakes in the file. I would appreciate it a lot if I am informed on any unusual behavior of the data file so that I can have the chance to correct, if any.

Once everything is validated, I will deliver the codes, and python scripts for you to fully process MODIS and SeaWifs LUT in your computer environment.

For OCI, I need to write a whole new set of scripts, so it will take a longer time.

Hope you all have a great day,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department,
UMBC
pwzhai@umbc.edu

On Mar 18, 2021, at 11:53 PM, Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov> wrote:

Pengwang,

This is the file I currently generate and provide to Zia's code to produce OCI LUTs. It contains the center wavelengths and the band-averaged Rayleigh Optical depth and depolarization.

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/afrt_input_oci.txt

I generate the above from interpolated spectral response functions, as provided below in ascii and hdf formats.

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/OCI_RSRs.txt

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/OCI_RSRs.hdf

The same files for SeaWiFS and MODIS/Aqua are here:

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/afrt_input_seawifs.txt

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/SeaWiFS_RSRs.txt

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/SeaWiFS_RSRs.hdf

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/afrt_input_modisa.txt

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/MODISA_RSRs.txt

https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/MODISA_RSRs.hdf

The full suite of Rayleigh and aerosol LUTs for OCI, as currently produced, are here:
https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/tmp/ocis/

Also, just for your information, this is a sampling of the sensors for which we currently produce (or have produced) AC LUTs:
https://oceancolor.gsfc.nasa.gov/filesshare/bryan_franz/bandpass/

It also shows some of the other band-integrated quantities we need for the AC algorithm. One of the advantages of pre-computing the band-integrated quantities (e.g., Rayleigh optical thickness) is that I also need those values in the AC algorithm (e.g., to correct the Rayleigh reflectance for actual atmospheric pressure).

-- bryan

Bryan Franz
Asst. Chief for Science Research
Ocean Ecology Laboratory (616)
NASA Goddard Space Flight Center

On 3/18/21, 4:09 PM, "Pengwang Zhai"
<pwzhai@umbc.edu> wrote:

Thanks, Amir, for the slides. In particular, I will need the following data in the first step:

1. Aerosol Optical Depth values.
2. Solar and viewing zenith angles, as well as viewing azimuth angles. In my configuration, the solar azimuth angle is always 0, so that I only need the viewing azimuth angles relative to the solar azimuth angles.
3. SeaWifs and MODIS spectral response functions as well as center wavelengths.
4. I will take the wind speed values from your slides.

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department,
UMBC
pwzhai@umbc.edu

On Mar 18, 2021, at 3:54 PM, Ibrahim, Amir
(GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

Attached the presentation from today's discussion. We can provide you with more input details as needed. I still need to wrap my head around the separation of the direct and diffuse transmittance terms.

Thanks again!

Amir

From: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Sent: Thursday, March 18, 2021 11:02 AM

To: Pengwang Zhai <pwzhai@umbc.edu>

Cc: Franz, Bryan A. (GSFC-6160)

<bryan.a.franz@nasa.gov>; Werdell, Jeremy

(GSFC-6160) <jeremy.werdell@nasa.gov>;

Knobelspiesse, Kirk D. (GSFC-6160)

<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng

(GSFC-616.0)[SCIENCE SYSTEMS AND

APPLICATIONS INC] <meng.gao@nasa.gov>;

BAILEY, SEAN W. (GSFC-6160)

<sean.w.bailey@nasa.gov>; Ahmad, Ziauddin

(GSFC-616.0)[Science Application

International Corp.] <ziauddin.ahmad-

1@nasa.gov>

Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Dear all,

This is a reminder that we will meet today
at 2 pm. Our agenda is:

- Pengwang's update on his RT development (15 min).
- Atmospheric Correction LUTs generation and description (15 min).
- Discussion and path forward (30 min).

Looking forward!

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>

Sent: Friday, March 12, 2021 11:04 AM

To: Ibrahim, Amir (GSFC-6160)

<amir.ibrahim@nasa.gov>

Cc: Franz. Brvan A. (GSFC-6160)

<bryan.a.franz@nasa.gov>; Werdell, Jeremy
(GSFC-6160) <jeremy.werdell@nasa.gov>;
Knobelspiesse, Kirk D. (GSFC-6160)
<kirk.d.knobelspiesse@nasa.gov>; Gao, Meng
(GSFC-616.0)[SCIENCE SYSTEMS AND
APPLICATIONS INC] <meng.gao@nasa.gov>;
BAILEY, SEAN W. (GSFC-6160)
<sean.w.bailey@nasa.gov>
Subject: Re: [EXTERNAL] Radiative Transfer
integration into OBPG

Hello, Amir,

I will update with you guys on the new
developments. I am not sure how we handle
versions. Let's discuss this and I will
listen to your comments on this. Also I
will listen to all the topics you
mentioned, and in particular what you need
and do any necessary development and
revision to the code to make it easier for
generating the LUT.

Take care,

Yours,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department,
UMBC
pwzhai@umbc.edu

On Mar 11, 2021, at 2:35 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Great. Thanks, Pengwang. If you would like, you can give us a 15 min update to your latest developments and how you handle versioning. We would like to keep the discussion at high level in order to better understand the necessary steps required to transition our processing using your code. For example, we need to discuss long term support, documentation, your code computing requirements and how that match our requirements and constraints. We also need to set a plan for a pilot example to generate LUTs and formatting needed in the processing code. I will refine my points here with help from others this week.

Amir

From: Pengwang Zhai <pwzhai@umbc.edu>
Sent: Thursday, March 11, 2021 1:40 PM
To: Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov>
Cc: Franz, Bryan A. (GSFC-6160) <bryan.a.franz@nasa.gov>; Werdell, Jeremy (GSFC-6160) <jeremy.werdell@nasa.gov>; Knobelspiesse, Kirk D. (GSFC-6160) <kirk.d.knobelspiesse@nasa.gov>; Gao, Meng (GSFC-616.0) [SCIENCE SYSTEMS AND APPLICATIONS INC] <meng.gao@nasa.gov>; BAILEY, SEAN W. (GSFC-6160) <sean.w.bailey@nasa.gov>
Subject: [EXTERNAL] Re: Radiative Transfer integration into OBPG

Hello, Amir,

Thanks for reaching out on this. 18th at 2 pm is a good time for me. I have accepted your meeting invitation.

Since my last deliver, I have made some significant changes to the simulator package, including a phase matrix simulation package to be used for the simulator, more options for UV bio optical models, and some other code maintenance. I will update with you guys once I manage to do some more testing on the code.

Let me know what I need to prepare before the meeting,

Take care,

Pengwang

Pengwang Zhai (he/his/him)
Graduate Program Director, Atmospheric
Physics,
Associate Professor, Physics Department,
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pwzhai@umbc.edu

On Mar 11, 2021, at 1:28 PM, Ibrahim, Amir (GSFC-6160) <amir.ibrahim@nasa.gov> wrote:

Hi Pengwang,

We would like to have a conversation with you regarding the possibility of integrating your radiative transfer code into our data processing system. We would like to use your code to generate the atmospheric correction LUTs for OCI and for all heritage missions. Your code would be valuable to us. Saying that, we also need to discuss certain basic requirements in order to transition your code into ODPS. We also would want to understand your needs/support in order to move forward with this step.

We are available Thursday the 18th at 2 pm, but if you are not available, please suggest other dates. I will send some discussion points before the meeting.

Thank you and we are looking forward to this discussion.

Regards,
Amir

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<Mail Attachment.ics>

<LUT_RT.pptx>

<PWZ_LUT_v1.pdf>

<foam.f>

<aerosol_model1_aot_0.1.pdf>

<f80_r70.png>

